

Week 8 Project Summary — Final Closure

AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track

1. Project Journey Overview

Over Weeks 4-8, this project moved from problem understanding (Week 4), through initial analysis and KPI development (Week 5), advanced segment-level validation and cross-track integration (Week 6), independent testing and refinement (Week 7), to this final, presentation-ready analytics package (Week 8).

2. What Was Completed in Week 8

Finalised the 5 core KPIs and two headline findings as the confirmed, tested analytical foundation for HealthConnect's no-show problem. Compiled the full Weeks 5-7 body of work into one final Analytics & Decision Support Package, including an executive summary for non-technical stakeholders, final business recommendations, and a documented final cross-track contribution to the Data Science track.

3. Final Key Findings

- Previous no-show history is the strongest, most reliable predictor of future no-shows (43.5% to 68.8% as prior no-shows increase), confirmed valid across every age group and by independent recalculation.
- Distance to the clinic is a real, secondary factor, especially for patients without an existing no-show pattern, though the relationship is a general trend rather than strictly linear.
- Reminders and waiting time show weak-to-no effect and should not be the focus of intervention efforts.

4. Cross-Track Collaboration (Final)

Collaborated with the Data Science track by delivering a final, validated, ranked feature recommendation for their no-show prediction model, including an honest caveat on the distance feature's non-linearity — ensuring their model is built on a tested analytical foundation rather than an assumed one.

5. Major Challenges Across the Project

The most significant technical challenge was building and debugging the derived calculation linking a patient's waiting time to their next appointment's outcome in Power BI (Week 5-6), which required resolving a circular reference error and a column-vs-measure aggregation error. The most significant analytical challenge was discovering, in Week 6, that distance and previous no-show history are not independent factors — requiring a combined-variable analysis rather than treating each finding in isolation.

6. Important Decisions Made

- Chose to recommend a distance x prior-history interaction feature to Data Science rather than treating the two variables as independent inputs, based on the Week 6 combined-effect finding.
- Chose to document the Week 7 distance non-linearity finding honestly as a caveat rather than ignoring it or overstating it as a failed validation.

7. Remaining Limitations

- The dataset is fictional and would require validation against real HealthConnect data before operational deployment.
- The 3+ previous no-shows segment remains based on a small sample size.
- This analysis identifies correlational patterns, not proven causation; recommendations should be piloted before full rollout.

8. Contribution to the Overall HealthConnect Project

The Data Analytics track's work provided the evidence-based foundation the entire HealthConnect solution rests on: identifying which patient and appointment factors genuinely predict no-shows, validating those findings through independent testing, and translating them into both a stakeholder-ready business case and a technically validated feature set for the Data Science track's predictive model.

9. Reflection — Lessons Learned

The most valuable lesson from this project was learning that a finding being true 'on average' does not mean it holds true everywhere — the Week 6 discovery that distance and prior no-show history interact, and the Week 7 discovery of a minor non-linearity in the distance relationship, both came from deliberately stress-testing initial conclusions rather than accepting them at face value. This is a core analytical habit: always ask whether a pattern survives being looked at from a different angle before relying on it for a real decision.